

Lab 4 plus

*Lab 4 plus is a combination of Lab 4 and an extra activity on ARP.

Packet Tracer Simulation – Exploration of ARP and Switch Table Communications

Objectives

- To explore ARP and switching operations.

Introduction

The topology is given to you. All IP addresses have been assigned to all devices. Please follow each step in sequence.

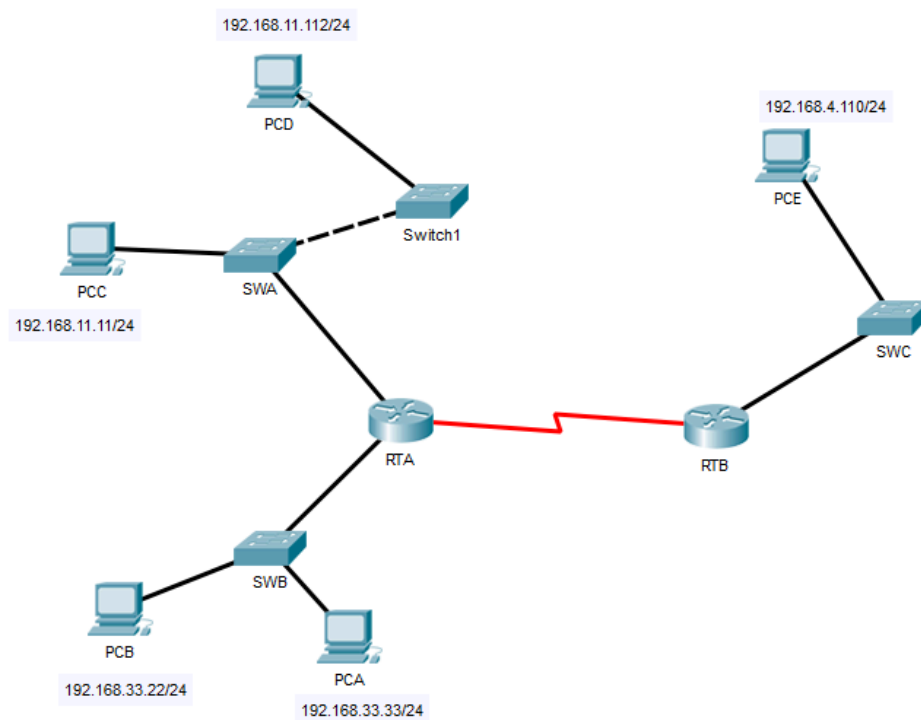


Figure 1

Part 1: Review the topology

Step 1: Open the *Lab_4_ARP.pkt*, and Perform the following tasks.

- a. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol Address           Age (min)  Hardware Addr  Type   Interface
Internet 192.168.11.1             -    0002.4A00.0E91  ARPA   FastEthernet1/0
Internet 192.168.33.1             -    000C.CF0C.593A  ARPA   FastEthernet0/0
RTA#
```

- b. At Router RTB, enter the CLI. At the command prompt type the commands as in Figure 2. Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
Protocol Address           Age (min)  Hardware Addr  Type   Interface
Internet 192.168.4.1             -    0001.977A.B614  ARPA   FastEthernet0/0
RTB#
```

- c. At Switches SWA, SWAB and SWC, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp
SWA#show mac-address-table
```

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       0002.4a00.0e91   DYNAMIC   Fa0/1
1       000c.8546.7d85   DYNAMIC   Fa1/1
SWA#
```

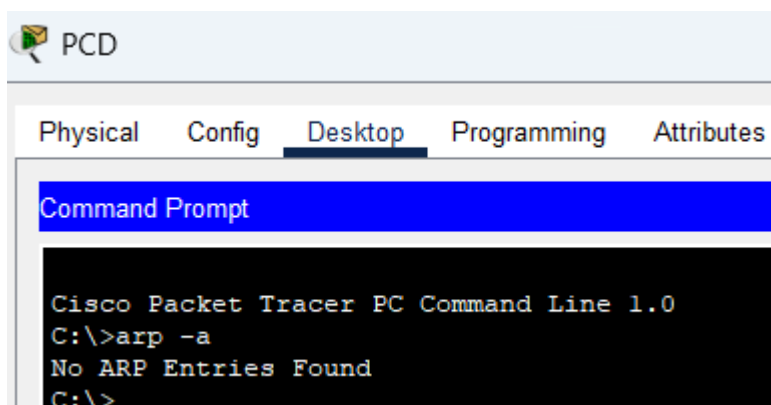
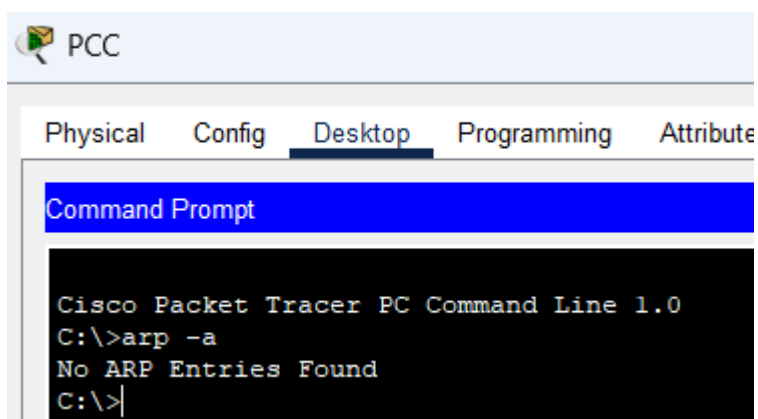
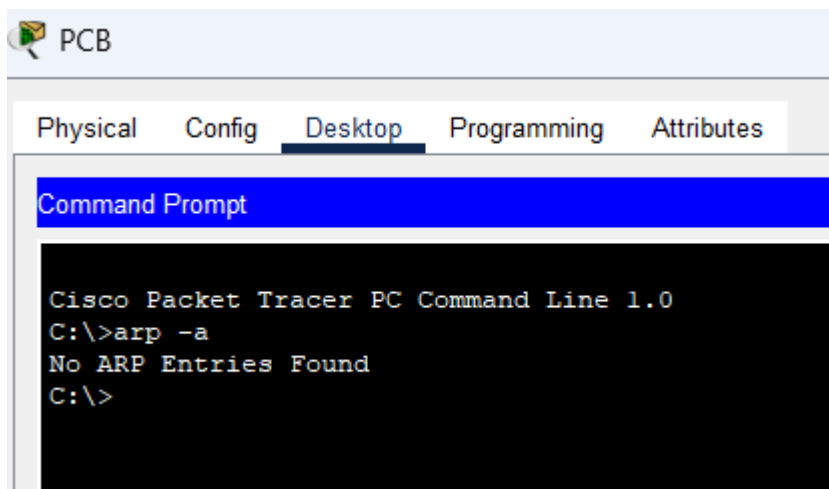
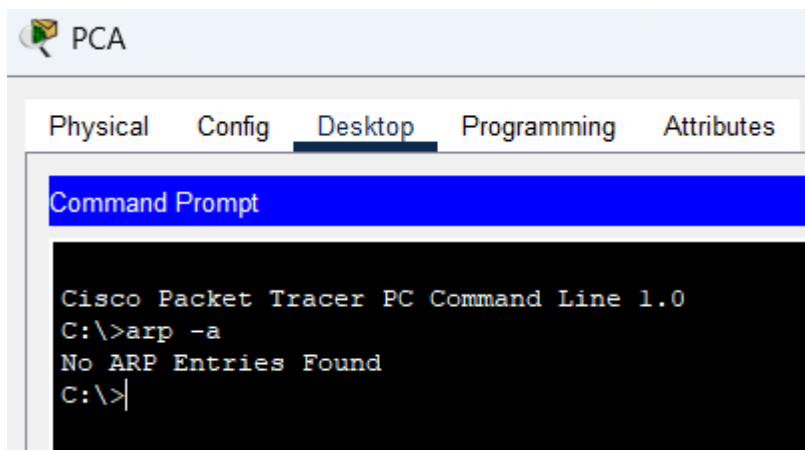
```
SWB>enable
SWB#show arp

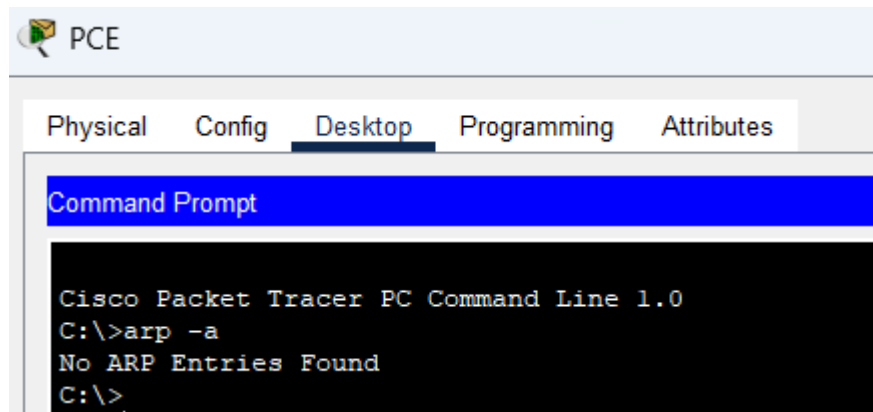
SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       000c.cf0c.593a   DYNAMIC   Fa0/1
SWB#
```

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       0001.977a.b614   DYNAMIC   Fa0/1
SWC#
```

- d. At PCA, click on the PC icon, and then choose Desktop-Command Prompt. At the command prompt type **arp -a** and click enter. Snap the results after the last command and paste it here. Do this to all PCs in the topology.





- e. What are your thoughts on the results?

The results show that ARP tables start empty, switches use MAC tables instead, and routers need ARP to find MAC addresses for IPs.

Part 2: Generate Network Traffic

Step 1: Generate traffic between PCA and PCB.

In the command prompt Perform the following tasks task to reduce the amount of network traffic viewed in the simulation.

- Click **PCA** and click the Desktop tab > Command Prompt.
- Enter the **ping 192.168.33.22** command. This may take a few seconds.
- In the Command prompt of PCA, type **arp -a**. Paste the result of this command here.

```
C:\>arp -a
Internet Address      Physical Address      Type
192.168.33.22         0060.47ea.a746        dynamic
```

- In the Command prompt of PCB, type **arp -a**. Paste the result of this command here

```
C:\>arp -a
Internet Address      Physical Address      Type
192.168.33.33         0002.1755.9a06        dynamic
```

- In the Command prompt of PCC, PCD abd PCE, type **arp -a**. Paste the result of this command here.

PCC:

```
C:\>arp -a
No ARP Entries Found
```

PCD:

```
C:\>arp -a
No ARP Entries Found
C:\>
```

PCE:

```
C:\>arp -a
No ARP Entries Found
C:\>
```

Step 2: Generate traffic between PCC to all other PC except PCA.

- Click **PCC** and click the Desktop tab > Command Prompt.
- Enter the **ping 192.168.33.22** command (ping to PCB). Then type **arp -a**. Paste the result after these commands here.

```

C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Request timed out.
Reply from 192.168.33.22: bytes=32 time<lms TTL=127
Reply from 192.168.33.22: bytes=32 time<lms TTL=127
Reply from 192.168.33.22: bytes=32 time<lms TTL=127

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a

```

Internet Address	Physical Address	Type
192.168.11.1	0002.4a00.0e91	dynamic

- c. Enter the **ping 192.168.11.112** command (ping to PCD). Then type **arp -a**. Paste the result after these commands here.

```

C:\>ping 192.168.11.112

Pinging 192.168.11.112 with 32 bytes of data:

Reply from 192.168.11.112: bytes=32 time<lms TTL=128
Reply from 192.168.11.112: bytes=32 time<lms TTL=128
Reply from 192.168.11.112: bytes=32 time<lms TTL=128
Reply from 192.168.11.112: bytes=32 time<lms TTL=128

Ping statistics for 192.168.11.112:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a

```

Internet Address	Physical Address	Type
192.168.11.1	0002.4a00.0e91	dynamic
192.168.11.112	0001.6462.0278	dynamic

- d. Enter the **ping 192.168.4.110** command (ping to PCE). Then type **arp -a**. Paste the result after these commands here.

```

C:\>ping 192.168.4.110

Pinging 192.168.4.110 with 32 bytes of data:

Request timed out.
Reply from 192.168.4.110: bytes=32 time=1ms TTL=126
Reply from 192.168.4.110: bytes=32 time=4ms TTL=126
Reply from 192.168.4.110: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.4.110:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 4ms, Average = 2ms

C:\>arp -a

```

Internet Address	Physical Address	Type
192.168.11.1	0002.4a00.0e91	dynamic
192.168.11.112	0001.6462.0278	dynamic

- e. Discuss the results you got from all the commands on PCC.

PCC only learns MAC addresses of devices in the same network in the same router. For other networks, it only knows the router's MAC address, and the router handles the rest.

- f. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```

RTA>enable
RTA#show arp

```

```

RTA>enable
RTA#show arp

```

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	192.168.11.1	-	0002.4A00.0E91	ARPA	FastEthernet1/0
Internet	192.168.11.11	6	00D0.D39A.C0D9	ARPA	FastEthernet1/0
Internet	192.168.33.1	-	000C.CF0C.593A	ARPA	FastEthernet0/0
Internet	192.168.33.22	6	0060.47EA.A746	ARPA	FastEthernet0/0

- g. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```

RTB>enable
RTB#show arp

```



```

RTB>enable
RTB#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.4.1 - 0001.977A.B614 ARPA FastEthernet0/0
Internet 192.168.4.110 5 0060.702D.7C08 ARPA FastEthernet0/0

```

Step 3: Switch MAC address table.

- a. At Switch SWA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```

SWA>enable
SWA#show arp

SWA#show mac-address-table

```

```

SWA>enable
SWA#show arp

SWA#show mac-address-table
      Mac Address Table
-----
Vlan  Mac Address      Type      Ports
----  -
1     0002.4a00.0e91    DYNAMIC   Fa0/1
1     000c.8546.7d85    DYNAMIC   Fa1/1
SWA#

```

- b. At Switch SWB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```

SWB>enable
SWB#show arp

SWB#show mac-address-table

```

```

SWB>enable
SWB#show arp

SWB#show mac-address-table
      Mac Address Table
-----
Vlan  Mac Address      Type      Ports
----  -
1     000c.cf0c.593a    DYNAMIC   Fa0/1
SWB#

```

- c. At Switches SWC and Switch1, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
```

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
      1    0001.977a.b614    DYNAMIC Fa0/1
SWC#|
```

- d. Do switches use arp table? (Y/N)

No

- e. Explain your answer in (d) **Hint: the answer may surprise you. Google for the explanation. It is not part of NetComm syllabi, it is just for knowledge..*

Switches do not use ARP tables because they operate at Layer 2 (Data Link Layer) of the OSI model, while ARP works at Layer 3 (Network Layer). Instead of ARP, switches use a MAC address table (CAM table) to map MAC addresses to specific switch ports.

- f. What information does the command `show mac-address-table` gives?

MAC addresses learned by the switch

Part 3: Attach wireless lab results.

In this part, you will use **Lab_4_Wireless.pka** file.

Step1: Change the filename of the pka file.

- Change the Lab 4 filename to include your name. **Example: Lab4AliAhmad.pkt*
- Go through the instructions. As you complete the tasks, you will see the bottom right hand corner of the pkt file increase in completion percentage, until you get 100/100.

The screenshot displays the Packet Tracer interface. On the left, a network topology is shown with a central switch S1 connected to three PCs (PC1, PC2, PC3) and a wireless router WRS2. PC1 is connected to S1 F0/11 (172.17.10.21/24, VLAN 10). PC2 is connected to S1 F0/18 (172.17.20.22/24, VLAN 20). PC3 is connected to WRS2 (172.17.40.100/24). WRS2 is connected to S1 G0/1 (172.17.88.25/24, VLAN 88). A subinterface list on the left shows: G0/0.10 172.17.10.1/24, G0/0.20 172.17.20.1/24, and G0/0.88 172.17.88.1/24.

On the right, the 'Packet Tracer – Configuring Wireless LAN Access' activity wizard is open. It shows an 'Addressing Table' with the following data:

Device	Interface	IP Address	Subnet Mask	Default Gate
R1	G0/0.10	172.17.10.1	255.255.255.0	N/A
	G0/0.20	172.17.20.1	255.255.255.0	N/A
	G0/0.88	172.17.88.1	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.1

At the bottom of the wizard, the 'Completion' status is shown as '100/100' with a progress bar and a '1/1' indicator, all enclosed in a red box.

- Once you have completed fully, capture the screen (which includes the filename, the topology and the activity wizard showing completion) and paste it here.

Cisco Packet Tracer - C:\Users\Owner\Desktop\Lab 4 NETCOM\LabAliffathi.pka - Guest - 2025-01-30 18:48:08

File Edit Options View Tools Extensions Window Help

Logical Physical x 4, y: 505

Configure the DHCP server.

c. Scroll to the bottom of the page and click **Save Settings**.

Step 4: Configure wireless access and security.

- At the top of the window, click **Wireless**. Set the **Network Mode** to **Wireless-N Only** and change the SSID to **WRS_LAN**.
- Disable **SSID Broadcast** and click **Save Settings**.
- Click the **Wireless Security** option.
- Change the **Security Mode** from **Disabled** to **WPA2 Personal**.
- Configure **cisco123** as the passphrase.
- Scroll to the bottom of the page and click **Save Settings**.

Part 2: Configure a Wireless Client

Step 1: Configure PC3 for wireless connectivity.

Because SSID broadcast is disabled, you must manually configure PC3 with the correct SSID and passphrase to establish a connection with the router.

- Click **PC3 > Desktop > PC Wireless**.
- Click the **Profiles** tab.
- Click **New**.
- Name the new profile **Wireless Access**.
- On the next screen, click **Advanced Setup**. Then manually enter the SSID of **WRS_LAN** on **Wireless Network Name**. Click **Next**.
- Choose **Obtain network settings automatically (DHCP)** as the network settings, and then click **Next**.
- On **Wireless Security**, choose **WPA2-Personal** as the method of encryption and click **Next**.
- Enter the passphrase **cisco123** and click **Next**.
- Click **Save** and then click **Connect to Network**.

Step 2: Verify PC3 wireless connectivity and IP addressing configuration.

- The **Signal Strength** and **Link Quality** indicators should show that you have a strong signal.
- Click **More Information** to see details of the connection including IP addressing information.
- Close the **PC Wireless** configuration window.

Part 3: Verify Connectivity

All the PCs should have connectivity with one another.

Time Elapsed: 00:11:20 Completion: 100/100

☒ Dock 1/1

Time: 00:11:12

4331 4321 1941 2901 2911 81910M 81940M 829 1240 PFRouter PFRempty 1841 26200M 26210M 2

2621XM

Scenario 0

Fire	Last Status	Source	Destination	Type	Color
	--	PC3	PC2	ICMP	
	--	PC3	PC1	ICMP	

Cisco Packet Tracer - C:\Users\Owner\Desktop\Lab 4 NETCOM\LabAliffFathi.pka - Guest - 2025-01-30 18:48:08

File Edit Options View Tools Extensions Window Help

Activity Results Time Elapsed: 00:14:16

Congratulations Guest! You completed the activity.

Overall Feedback **Assessment Items** Connectivity Tests

Expand/Collapse All Show Only Incorrect Item

Assessment Items	Status	Points
Network		
PC3		
Wireless		
Security Mode		
Authen Type	Correct	1
Pass Phrase	Correct	4
SSID	Correct	5
WRS2		
(deprecated) DHCP Server		
(deprecated) DHCP Enable	Correct	10
(deprecated) Pools		0
(deprecated) Pool linksysPool		0
(deprecated) Default Gateway	Correct	10
Default Gateway	Correct	10
Ports		
Internet		
IP Address	Correct	10
Link to S1		
Connects to FastEthernet0/7	Correct	5
Type	Correct	5
Wireless		0
Wireless		
Security Mode		
Authen Type	Correct	10
Pass Phrase	Correct	10
SSID	Correct	10
SSID BroadCast	Correct	10

Score : 100/100

Item Count : 13/13

Component	Items/Total	Score
Device Connection	2/2	10/10
Wireless Client Configuration	3/3	10/10
Wireless Router Configuration	8/8	80/80

Close